Author: In practice this is the method generally used. However one should note that in this case we have assumed, first, that $\lim_{n\to\infty} \frac{1}{n} = 0$, and, second, the validity of the following rules: $\boxed{\lim_{n\to\infty} \frac{x_n}{y_n} = \frac{\lim_{n\to\infty} x_n}{\lim_{n\to\infty} y_n}}$ (2) and

$$\boxed{\lim_{n\to\infty}(x_n+z_n) = \lim_{n\to\infty}(x_n) + \lim_{n\to\infty}(z_n)} \quad (3)$$ where

where $x_n = 1$, $y_n = 1+\frac{1}{n}$, and $z_n = \frac{1}{n}$. Later on we shall discuss these rules, but at this juncture I suggest that we simply use them to compute several limits. Let us discuss two examples.

Example 1: Find $\lim_{n\to\infty} \frac{3n-1}{5n-6}$

San: $$\lim_{n\to\infty} \frac{3n-1}{5n-6} = \frac{\lim_{n\to\infty}(3n-1)}{\lim_{n\to\infty}(5n-6)} = \frac{\lim_{n\to\infty} \frac{3n-1}{n}}{\lim_{n\to\infty} \frac{5n-6}{n}}$$

$$= \frac{\lim_{n\to\infty}\left(3-\frac{1}{n}\right)}{\lim_{n\to\infty}\left(5-\frac{6}{n}\right)} = \frac{3}{5}$$

Author: O.K. **Example 2:** Compute $\lim_{n\to\infty} \frac{6n^2-1}{5n^2+2n-1}$

San: $$\lim_{n\to\infty} \frac{\frac{6n^2-1}{n^2}}{\frac{5n^2+2n-1}{n^2}} = \boxed{\lim_{n\to\infty} \frac{6-\frac{1}{n^2}}{5+\frac{2}{n}-\frac{1}{n^2}} = \frac{6}{5}}$$

Author: Well, we have examined the concept of the limit of sequence. Moreover we have learnt a little how to calculate limits. Now it is time to discuss some properties of sequences with limits. Such sequences are called convergent.

San: Some resources for further reading

1. The History of the Calculus and its Conceptual Development — Carl B. Boyer

- Covers Isaac Newton and Gottfried Wilhelm Leibniz
- Explains infinitesimals (the "almost zero" idea)
- Shows how calculus worked before it was logically justified.